



AIMS: An Internet Media Store

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SOLID Design Principles

Each member will write a report and comment directly on the source code sections of the responsible use cases that violate SOLID design principles. Please name your file as the convention “StudentID_StudentName_SOLID.pdf” for submission.

The report must clearly specify:

- Which code section
- Which principle is violated (S, O, L, I, or D)?
- Why: Explain how future extensions or modifications would cause the design to violate that principle.
- Proposed solution direction: Present ideas to resolve the issue so that the principle is no longer violated. For example: Introduce a new parent class Xxx with method m(), and let class Yyy extend it and override method m().

Design Patterns

For the parts of the design or source code that are not well-structured (i.e., do not ensure low coupling and high cohesion, or violate design principles), perform the following tasks:

- A report (following the naming convention “StudentID_StudentName_DesignPatterns.pdf”) that clearly states:
 - How the design has been improved
 - Whether any design patterns were applied, for which requirement changes, and why
- The updated Astah Project (.astah) files and exported image files for the improved design (class diagram, sequence diagram)
- Refactor the source code to achieve clean code (ensuring high cohesion, low coupling, compliance with the SOLID principles, and proper application of design patterns). Commit and push the source code to GitHub regularly during the refactoring process.

Important Note: Your task is to **IMPROVE THE DESIGN** and **REFACTOR THE SOURCE CODE** to make the existing design better (easy to extend, easy to maintain) — not to design and implement new features. Therefore, you **are not** required to design or implement the new features listed below. These new features are only potential future requirements, serving as input for analyzing and improving the current design and source code.

1. Add a new non-media product

Currently, AIMS sells various types of media products, including books, newspapers, CDs, and DVDs.

In the future, as the store expands, other types of products may be considered for sale, such as clothing (which may have specific attributes like size, fabric material, etc.).

2. Add a new media

Currently, media products integrated with artificial intelligence (AI) are becoming increasingly popular in the market. Among them are audiobooks that use AI technology to convert text into natural-sounding speech, personalize content according to user preferences, and support intelligent interaction.

In the future, a new product named **E-Book Reader** is introduced. This product integrates artificial intelligence features to enhance the reading experience. The specific attributes of this product include hardware specifications and AI-related capabilities such as text summarization, AI translation, and text-to-speech functionality. The information for the new product type is as follows:

- Screen Size (*e.g., 6.8 inches E-Ink display*)
- Storage Capacity (*e.g., 16GB*)
- Battery Life (*e.g., up to 10 weeks per charge*)
- Supported Formats (*e.g., PDF, EPUB, MOBI*)
- AI Features (*e.g., Text Summarization, AI Translation, Smart Highlighting*)
- Text-to-Speech Support (*e.g., Yes – AI natural voice synthesis*)
- Supported Languages for AI Processing (*e.g., English, Vietnamese, Japanese*)

3. Change the shipping fee calculation strategy

Currently, the shipping fee is calculated based on the total weight of the items in the order.

In the future, other shipping fee calculation strategies will be applied. For example, the shipping fee will be calculated based on the actual weight and the volumetric weight of the package. The volumetric weight is determined by the package dimensions: length, width, and height, using the following formula:

Volumetric weight (kg) = Length (cm) × Width (cm) × Height (cm) / 6000

The weight used to calculate the shipping fee is:

Chargeable weight = MAX (actual weight, volumetric weight)

Example: A package weighs 2 kg and has dimensions $50 \times 60 \times 10$ (cm³), volumetric weight = $50 \times 60 \times 10 / 6000 = 5$ kg. Therefore, Chargeable weight = MAX (2, 5) = 5 kg.

The shipping fee is then calculated according to the AIMS specification.

4. Integrate an additional payment gateway

Currently, AIMS integrates VietQR for QR code payments and PayPal for card payments.

To increase flexibility and provide more payment options for customers, an additional payment gateway (such as VNPay, Momo, ZaloPay, etc.) should be integrated into the system. This will allow the store to support more diverse payment methods, such as other e-wallets, banking gateways, or international payment providers, depending on business needs in the future.

5. Change the notification delivery strategy

Currently, AIMS sends notifications only via email.

To enhance flexibility and improve user experience, additional notification delivery channels should be introduced. In the future, notifications may also be sent through:

- SMS messages
- Zalo messages
- Push notifications

By supporting multiple delivery methods, the system can provide faster and more convenient communication with users across different platforms.